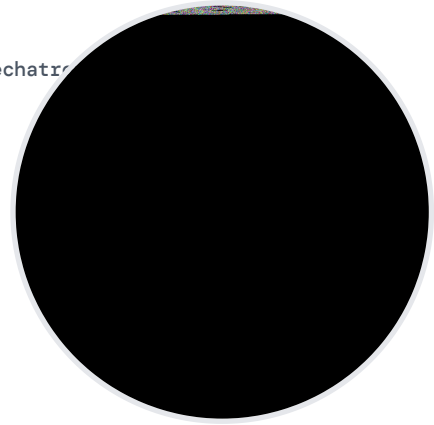
PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatr



Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

View Projects | Curriculum Vitae

Email | LinkedIn | WhatsApp

Mechanical
3D CAD modeling, mechanism design, and fabrication.

Electronics
Schematic design, circuit wiring, and hardware assembly.

Embedded Control
Microcontrollers, sensors, and actuator integration.

Prototyping
Physical build, functional testing, and validation workflows.

01 / PROJECTS

Engineering work, in practice.

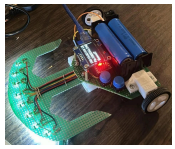
A collection of 5 projects across undergraduate research, national robotics competition, capstone design, and independent builds.

01 / FINAL PROJECT

02 / CAPSTONE DESIGN

03 / KONTES ROBOT ABU INDONESIA (KRAI)

04 / PROYEK MANDIRI



05 / PROYEK MANDIRI

Robot Manipulator 5-DOF

5-DOF articulated robotic arm prototype powered by an STM32F411 microcontroller and 3D-printed cycloidal gearboxes. Integrates AS5600 magnetic encoders in a closed loop to reduce backlash, achieving 79.8% accuracy across 250–1,000 g loads.



Click to view details

Pendingin Panel Surya Otomatis

Automated water spray cooling system to lower solar panel surface temperature driven by Arduino Uno. Uses temperature sensing and relay control to trigger a submersible pump above >30°C and shut down below <25°C to restore efficiency.



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Kompetisi Robot Basket

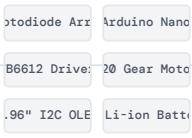
Mobile basketball robot for Kontes Robot ABU Indonesia (KRAI). Served as a core mechanical team member designing the shooter, dribbler, and drive wheel configuration, advancing to the National Stage at ITB Jatinangor.



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Robot Line Follower

Autonomous line-tracking robot featuring differential drive steering. Equipped with an 8-sensor bar built from LED and photodiode pairs, an Arduino Nano microcontroller, a TB6612FNG driver, and micro N20 metal gear motors.



Click to view details

Robot Sumo

1 kg competition-class sumo robot (20 × 20 cm) built on an acrylic chassis with a front aluminum scoop. Delivers 4 kg push force via 4 PG24 planetary gear motors, steered wirelessly from a smartphone via Bluetooth connection.



Click to view details

02

Hands-on practice in manufacturing environments.

Two internship tenures at PT Promanufacture Indonesia (Formulatrix), completed during vocational school in 2021 in Mechanical Engineering and during university in 2025 in Process Engineering.

FORMULATRIX®

PT Promanufacture Indonesia

Hands-on CAD modeling, workshop fabrication, and assembly technical documentation in a manufacturing environment.

Process Engineering Intern

Jul 2025 – Sep 2025 · 3 mos

SALATIGA, CENTRAL JAVA

- Gained familiarity with SMT line workflows and prepared assembly and wiring Work Instructions (WI) for the production team.
- Designed PCB solder paste stencil panels in Easy-PC and exported Gerber files for vendor tooling.
- Designed production support brackets in Onshape and fabricated parts using 3D printing.

Mechanical Engineering Intern

May 2021 – Oct 2021 · 6 mos

SEMARANG, CENTRAL JAVA

- Designed 3D components in Onshape for laboratory automation equipment in compliance with workshop manufacturing requirements.
- Prepared CNC machining toolpaths using OneCNC software.
- Executed hands-on fabrication and assembly: 3D printing, deburring, filing, grinding, sheet metal bending, and mechanical assembly.

03 / SKILLS

Technical skills & engineering tools.

Software and tools with direct hands-on experience in academic research, industrial internships, and robotics engineering.

01 / MECHANICAL

CAD Modeling & Fabrication

- 3D parametric CAD modeling, assemblies & engineering drawings
- Mechanical transmission & custom Cycloidal Gearbox design
- Rapid mechanical prototyping & FDM 3D printing workflows
- Workshop bench fabrication, manual hand tools & power tools

SOFTWARE & TOOLS

Onshape · Fusion 360 · Inventor · Bambu Lab · Ultimaker Cura · Creality · OneCNC

02 / ELECTRONICS

PCB Design & Wiring

- Schematic capture design & multilayer PCB routing layout
- Electronic circuit simulation & virtual instrument testing
- Wiring harness layout, sensor cabling & pinout integration
- SMT line workflows (solder stencil, pick-and-place & reflow)

SOFTWARE & TOOLS

EasyEDA · Eagle · Proteus · Easy-PC

03 / EMBEDDED SYSTEMS

Firmware & Control

- Microcontroller firmware programming (STM32, ESP32, Arduino)
- Real-time closed-loop control systems & sensor feedback
- Serial communication protocols & I2C bus multiplexing
- Analytical kinematics modeling & motion trajectory planning

SOFTWARE & TOOLS

STM32 · Arduino · VS Code · C/C++ · Python

04 / ENGINEERING PRODUCTIVITY

Collaboration & Documentation

- Technical documentation authoring, test reports & thesis book
- Engineering data analysis, calculation sheets & sensor test logs
- System workflow modeling (flowcharts & engineering blocks)
- Standardized production documents (SOP & Work Instructions)

SOFTWARE & TOOLS

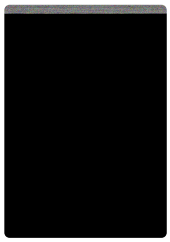
Google Docs · Google Sheets · Google Drive · Google Chat · Google Workspace · Miro

04 / CERTIFICATIONS & CREDENTIALS

Verified Qualifications & Certificates

Official competency certificates and tournament records.

INDUSTRY CERTIFICATION — Nov 2021



Pneumatic System Maintenance Technician

Credential ID: 62372/21/1185 · Unlimited Validity

Competency certificate from PersCert TÜV and FESTO Didactic validating technical skills in industrial pneumatic maintenance, valve circuits, and electro-pneumatic systems.

Click to view certificate

NATIONAL ROBOTICS TOURNAMENT — 7-8 Jul 2025



National Finalist - ABU Robocon Indonesia (KRAI 2025)

Belmawa Kemdiktisaintek & ITB · Team ROOBICS

Official certificate of National Finalist representing Telkom University (Team ROOBICS) in the Kontes Robot ABU Indonesia (KRAI) National Stage at ITB Jatinangor. Certificate No: 5468HK/IT1.B04.4/KM/2025.

Click to view certificate

ENGLISH PROFICIENCY

Jun 2026



English Proficiency Test (EPrT)

Total Score: 450 (Listening: 47, Grammar: 41, Reading: 47)

English proficiency certificate issued by Telkom University Language Center validating working technical comprehension.

Click to view certificate

PROFESSIONAL DEVELOPMENT

2025



Tel-U Career Coach (STAR) - Intrapreneurship

Directorate of Student Affairs, Career & Alumni · Telkom University

Official certificate awarded upon completing the Career Coach STAR development program in workplace adaptability, team dynamics, and intrapreneurship.

Click to view certificate

05 / EDUCATION

Academic Education History

Formal education journey spanning vocational mechatronics engineering to a bachelor degree in electrical engineering.

Telkom University

Bachelor of Science in Electrical Engineering

UNDERGRADUATE DEGREE

Sep 2022 - Aug 2026 · Bandung, West Java

Completed the degree in 3 years and 11 months with 154 credits. Authored an undergraduate final project titled "Design of a Manipulator Arm for a Patient Service Robot in a Hospital", focusing on 3D-printed cycloidal gearbox transmissions and closed-loop AS5600 magnetic sensing.

CUMULATIVE GPA	ACCREDITATION	TOTAL CREDITS
0.00 / 4.00	Unggul (BAN-PT) · IABEE	154 SKS Passed

[redacted]

SMK Negeri 7 Semarang

Mechatronics Engineering

4-YEAR PROGRAM · Jul 2018 - Jun 2022 · Semarang, Central Java

4-year technical mechatronics program with a 6-month industrial internship (PKL) in manufacturing and a 6-month capstone project in the 4th year assembling an industrial electro-pneumatic training kit.

GRADUATION SCORE	ACCREDITATION
0.00 / 100	Accredited "A" (BAN-S/M)

06 / ORGANIZATIONS

Organizational & Committee Experience

Active involvement in KRAI robotics research, PKKMB discipline management, and mentorship of cadreization participants at KMTE.

ROBOTICS RESEARCH GROUP

2024 - 2025

[redacted]

Electronics and Intelligence Robotic Research Group (EIRRG)

Mechanical Division · Team ROOBICS

Collaborated on CAD modeling, precision fabrication, and robot mechanism integration for Kontes Robot ABU Indonesia (KRAI 2025), qualifying for the National Finals at ITB Jatinangor.

STUDENT ORIENTATION COMMITTEE

2024

Panitia PKKMB Universitas Telkom

Discipline and Motivation Team (DMT)

Coordinated orientation protocols, crowd discipline, and field security for thousands of freshmen across faculties to ensure structured and orderly university induction events.

DEPARTMENT STUDENT GUILD

2023 - 2024

Keluarga Mahasiswa Teknik Elektro (KMTE)

Mentor Division · QUOTE 2023

Served as a mentor during the QUOTE 2023 cadreization program, guiding prospective members through organizational values, teamwork ethics, and departmental culture to join KMTE.

07 / GET IN TOUCH

Eager to learn, ready to contribute.

As a fresh electrical engineering graduate with a practical mechatronics background, I bring hands-on experience in prototyping and fabrication. I am eager to learn, adapt quickly, and be mentored by experienced engineering teams. Open to entry-level and internship engineering roles.

Curriculum Vitae

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LINKEDIN

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WHATSAPP

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